9

Operation of Ethernet systems

Objectives

When you have completed study of the chapter, you will have:

- Familiarized yourself with the 802 series of IEEE standards for networking
- Studied the details of the makeup of the data frames under DIX standard, IEEE 802.3, LLC, IEEE 802.1p, and IEEE 802.1Q
- Understood in depth how CSMA/CD operates for Half-Duplex transmissions
- Understood how multiplexing, Ethernet flow control, and PAUSE operations work
- Understood how full-duplex transmissions and auto-negotiation are carried out

9.1 Introduction

The OSI reference model was dealt with in chapter one wherein it was seen that the data link layer establishes communication between stations. It creates, transmits and receives data frames, recognizes link addresses, etc. It provides services for the various protocols at the network layer above it and uses the physical layer to transmit and receive messages. The data link layer creates packets appropriate for the network architecture being used. Network architectures (such as Ethernet, ARCnet, Token Ring, and FDDI) encompass the data link and physical layers

Ethernet standards include a number of systems that all fit within data link and physical layers. The IEEE has facilitated organizing of these systems by dividing these two OSI layers into sub layers, namely, the logical link control (LLC) and media access control (MAC) sub layers.

9.1.1 Logical link control (LLC) sublayer

The IEEE has defined the LLC layer in the IEEE 802.2 standard. This sub-layer provides a choice of three services viz. an unreliable datagram service, an acknowledgement datagram service, and a reliable connection oriented service. For acknowledged datagram or connection-oriented services, the data frames contain a source address, a destination address, a sequence number, an acknowledgement number, and a few miscellaneous bits. For unreliable datagram service, the sequence number and acknowledge number are omitted.

9.1.2 MAC sublayer

MAC layer protocols are a set of rules used to arbitrate access to the communication channel shared by several stations. The method of determining which station can send a message is critical and affects the efficiency of the LAN. Carrier sense multiple access/collision detection (CSMA/CD), the method used for Ethernet, will be discussed in this chapter.

It must be understood that medium access controls are only necessary where more than two nodes need to share the communication path, and separate paths for transmitting and receiving do not exist. Where only two nodes can communicate across separate paths, it is referred to as full-duplex operation and the need for arbitration does not exist.

To understand MAC protocols thoroughly, this chapter will look at the design of data frames, the rules for transmission of these frames, and how these affect design of Ethernet LANs. It was mentioned in chapter one that the original DIX standard and IEEE 802.3 standards differ only slightly. Both these standards will be compared in all aspects so as to make concepts clear and to prevent any confusion. We shall also see how backward compatibility is maintained in the case of faster versions of Ethernet.

9.2 IEEE/ISO standards

The IEEE has been given the task of developing standards for LAN under the auspices of the IEEE 802 committee. Once a draft standard has been agreed and completed, it is passed to the International Standards Organization ISO for ratification. The corresponding ISO standard, which is generally accepted internationally, is given the same number as the IEEE committee, with the addition of an extra ‘8’ in front of the number i.e. IEEE 802 is equivalent to ISO 8802.

These IEEE committees, consisting of various technical, study and working groups, provide recommendations for various features within the networking field. Each committee is given a specific area of interest, and a separate sub-number to distinguish it. The main committees and the related standards are described below.

9.2.1 Internetworking

These standards are responsible for establishing the overall LAN architecture, including internetworking and management. They are a series of sub standards, which include:

802.1B – LAN management

802.1D – Local bridging

802.1p – This is part of 802.1D and provides support for traffic-class expediting (for traffic prioritization in a switching hub) and dynamic multicast filtering)to identify which ports to use when forwarding multicast packages)

802.1Q – This is a VLAN standard providing for a vendor-independent way of implementing VLANs

802.1E – System load protocol
802.1F – Guidelines for layer management standards
802.1G – Remote MAC bridges
802.1I – MAC bridges (FDDI supplement)

802.2 Logical link control

This is the interface between the network layer and the specific network environments at the physical layer. The IEEE has divided the data link layer in the OSI model into two sublayers – the media access MAC sublayer, and the logical link layer LLC. The logical link control protocol (LLC) is common for all IEEE 802 standard network types. This provides a common interface to the network layer of the protocol stack.

The protocol used at this sub layer is based on IBM's SDLC protocol, and can be used in three modes, or types:

- **Type 1** – Unacknowledged connectionless link service
- **Type 2** – Connection-oriented link service
- **Type 3** – Acknowledged connectionless link service, used in real time applications such as manufacturing control

802.3 CSMA/CD local area networks

The carrier-sense, multiple access with collision detection type LAN is commonly, but erroneously, known as an Ethernet LAN. The CSMA/CD relates to the shared medium access method, as does each of the next two standards.

The committee was originally constrained to only investigate LANs with a transmission speed not exceeding 20 Mbps. However, that constraint has now been removed and the committee has published standards for a 100 Mbps 'Fast Ethernet' system and a 1000 Mbps 'Gigabit Ethernet', and, a 10000 Mbps '10 Gigabit Ethernet' system.

802.3ad Link aggregation

This standard is complete and has been implemented. It provides a vendor-neutral way to operate Ethernet links in parallel.

802.3x Full-duplex

This is a full-duplex supplement providing for explicit flow control by optional MAC control and PAUSE frame mechanisms.

802.4 Token bus LANs

The other major access method for a shared medium is the use of a token. This is a type of data frame that a station must possess before it can transmit messages. The stations are connected to a passive bus, although the token logically passes around in a cyclic manner.

802.5 Token ring LANs

As in 802.4, data transmission can only occur when a station holds a token. The logical structure of the network wiring is in the form of a ring, and each message must cycle through each station connected to the ring.

802.6 Metropolitan area networks

This committee is responsible for defining the standards for MANs. It has recommended that a system known as Distributed Queue Data Bus DQDB be utilized as a MAN standard. The explanation of this standard is outside the scope of this manual. The committee is also investigating cable television interconnection to support data transfer.

802.7 Broadband LANs Technical Advisory Group (TAG)

This committee is charged with ensuring that broadband signaling as applied to the 802.3, 802.4 and 802.5 medium access control specifications remains consistent. Note that there is a discrepancy between IEEE 802.7 and ISO 8802.7. The latter is responsible for slotted ring LAN standardization.

802.8 Fiber optic LANs TAG

This is the fiber optic equivalent of the 802.7 broadband TAG. The committee is attempting to standardize physical compatibility with FDDI and synchronous optical networks (SONET). It is also investigating single mode fiber and multimode fiber architectures.

802.9 Integrated voice and data LANs

This committee has recently released a specification for Isochronous Ethernet as IEEE 802.9a. It provides a 6.144 Mbps voice service (96 channels at 64 kbps) multiplexed with 10 Mbps data on a single cable. It is designed for multimedia applications.

802.10 Secure LANs

Proposals for this standard included two methods to address the lack of security in the original specifications. These are:

A secure data exchange (SDE) sublayer that sits between the LLC and the MAC sublayers. There will be different SDEs for different systems, e.g. military and medical.

A secure interoperable LAN System (SILS) architecture. This will define system standards for secure LAN communications.

The standard has now been approved. The approved standards listed below provide IEEE 802 environments with:

- Security association management
- Key management (manual, KDC, and certificate based)
- Security labeling
- Security services (data confidentiality, connectionless integrity, data origin authentication and access control)

The Key Management Protocol (KMP) defined in Clause 3 of IEEE 802.10c is applicable to the secure data exchange (SDE) protocol contained in the standards and other security protocols.

Approved standards (available at http://standards.ieee.org/getieee802/) include:

- IEEE Standard for Interoperable LAN/MAN Security (SILS), IEEE Std 802.10-1998
- Key Management (Clause 3), IEEE Std 802.10c-1998 (supplement)
- Security Architecture Framework (Clause 1), IEEE Std 802.10a-1999 (supplement)

802.11 Wireless LANs

Some of the techniques being investigated by this group include spread spectrum signaling, indoor radio communications, and infrared links.

Several Task Groups are working on this standard. Details and corresponding status of the task groups are as follows:

- The MAC group has completed developing one common MAC for WLAN applications, in conjunction with PHY group. Their work is now part of the standard

- PHY group has developed three PHYs for WLAN applications using Infrared (IR), 2.4 GHz Frequency Hopping Spread Spectrum (FHSS), and 2.4 GHz Direct Sequence Spread Spectrum (DSSS). Work is complete and has been part of the standard since 1997
- The Tga group developed a PHY to operate in the newly allocated UNII band. Work is complete, and is published as 802.11a-1999
- The Tgb group developed a standard for a higher rate PHY in 2.4 GHz band. Work is complete and issued as a part of 802.11b-1999

802.12 Fast LANs

Two new 100 Mbps LAN standards were ratified by the IEEE in July 1995, IEEE 802.3u and IEEE 802.12. These new 100 Mbps standards were designed to provide an upgrade path for the many 10s of millions of 10BaseT and token ring users worldwide. Both of the new standards support installed customer premises cabling, existing LAN management and application software.

IEEE 802.3 and IEEE 802.12 have both initiated projects to develop gigabit per second LANs initially as higher speed backbones for the 100 Mbps systems.

Demand priority summary

Demand priority is a protocol that was developed for IEEE 802.12. It combines the best characteristics of Ethernet (simple, fast access) and token ring (strong control, collision avoidance, and deterministic delay). Control of a demand priority network is centered in the repeaters and is based on a request/grant handshake between the repeaters and their associated end nodes. Access to the network is granted by the repeater to requesting nodes in a cyclic round robin sequence. The round robin protocol has two levels of priority, normal and high priority. Within each priority level, selection of the next node to transmit is determined by its sequential location in the network rather than the time of its request. The demand priority protocol has been shown to be fair and deterministic. IEEE 802.12 transports IEEE 802.3, Ethernet and IEEE 802.5 frames.

Scalability of demand priority: burst mode

The demand priority MAC is not speed sensitive. As the data rate is increased, the network topologies remain unchanged. However, since Ethernet or token ring frame formats are used the efficiency of the network would decrease with increased data rate.

To counteract this a packet burst mode has been introduced into the demand priority protocol. Burst mode allows an end node to transmit multiple packets for a single grant.

The new packet burst mode may also be implemented at 100 megabits per second, as it is backwards compatible with the original MAC. Analysis indicates that minimum efficiencies of 80% are possible at 1062.5 MBaud.

Higher speed IEEE 802.12 physical layers

Higher speed IEEE 802.12 will leverage some of the physical layers and control signaling developed for a fiber channel. The fiber channel baud rates leveraged by IEEE 802.3 are 531 MBaud and 1062.5 MBaud.

Multimode fiber and short wavelength laser transceivers will be used to connect repeaters or switches separated by less than 500 m within buildings. Single mode fiber and laser transceivers operating at a wavelength of 1300 nm will be used for campus backbone links. It has been shown that FC-0 and FC-1 can be used to support an IEEE 802.12 MAC or switch port.

A new physical layer under development in IEEE 802.12 will support desktop connections at 531 MBaud over 100 m of UTP category 5. The physical layer will utilize

all pairs of a four pair cable. The proposed physical layer incorporates a new 8B3N code and provides a continuously available reverse channel for control. There is no requirement for echo cancellation, which simplifies the implementation. Potential for class B radiated emissions compliance has been demonstrated.

9.3 Ethernet frames

The term Ethernet originally referred to a LAN implementation standardized by Xerox, Digital, and Intel; the original DIX standard. The IEEE 802.3 group standardized operation of a CSMA/CD network that was functionally equivalent to the DIX II or 'Bluebook' Ethernet.

Data transmission speeds of 100 Mbps (the IEEE 802.3u standard, also called Fast Ethernet) and 1000 Mbps (the 802.3z standard, also called Gigabit Ethernet) have been achieved, and, these faster versions are also included in term 'Ethernet'.

When the IEEE was asked to develop standards for Ethernet, Token Ring, and other networking technologies, DIX Ethernet was already in use. The objective of the 802 committee was to develop standards and rules that would be generic to all types of LANs so that data could move from one type of network, say Ethernet, to other type, say token ring. This had potential for conflict with the existing DIX Ethernet implementations. The '802' committee was therefore careful to separate rules for the old and the new since it was recognized there would be a coexistence between DIX frames and IEEE 802.3 frames on the same LAN.

These are the reasons why there is a difference between DIX Ethernet and IEEE 802.3 frames. Despite the two types of frames, we generally refer to both as 'Ethernet' frames in the following text.

9.3.1 DIX and IEEE 802.3 frames

A frame is a packet comprising data bits. The packet format is fixed and the bits are arranged in sequence of groups of bytes (fields). The purpose of each field, its size, and its position in the sequence are all meaningful and predetermined. The fields are Preamble, Destination Address, Source Address, Type or Length, DATA/LLC, and Frame Check Sequence, in that order.

DIX and IEEE 802.3 frames are identical in terms of the number and length of fields. The only difference is in the contents of the fields and their interpretation by the stations which send and receive them. Ethernet interfaces therefore can send either of these frames.

Figure 9.1 schematically shows the DIX as well as IEEE 802.3 frame structures.

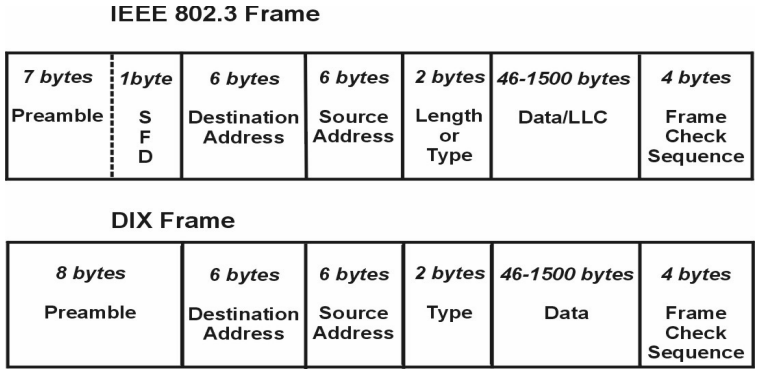


Figure 9.1
IEEE 802.3 and DIX frames

9.3.2 Preamble

The ‘preamble’ of the frame is like the introductory remarks of a speaker. If one misses a few words from a preamble being delivered by a speaker, one does not lose the substance of the speech. Similarly, the preamble in this case is used for synchronization and to protect the rest of the frame even if some start-up losses occur to the signal.

Fast and Gigabit Ethernet have other mechanisms for avoiding signal start-up losses, but in their frames a preamble is retained for purposes of backward compatibility.

Because of the synchronous communication method used, the preamble is necessary to enable all stations to synchronize their clocks. The Manchester encoding method used for 10 Mbps Ethernet is self-clocking since each list contains a signal transition in the module.

Preamble in DIX

Here the preamble consists of eight bytes of alternating ones and zeros, which appear as a square wave with Manchester encoding. The last two bits of the last byte are ‘1,1’. These ‘1,1’ bits signify to the receiving interface that the end of the preamble has occurred and that actual meaningful bits are about to start.

Preamble in IEEE 802.3

Here the preamble is divided in two parts, first one of seven bytes, and another of one byte. This one byte segment is called the start frame delimiter or SFD for short. Here, again, the last two bits of the SFD are ‘1,1’ and with the same purpose as in the DIX standard. There is no practical difference between the preambles of DIX and IEEE – the difference being only semantic.

9.3.3 Ethernet MAC addresses

These addresses are also called hardware addresses or media addresses. Each Ethernet interface needs a unique MAC address, and this is usually allocated at the time of manufacture. The first 24 bits of the MAC address consist of an organizationally unique identifier (OUI), in other words a ‘manufacturer ID’, assigned to a vendor by the IEEE. This is why they are also called vendor codes. The Ethernet vendor combines their 24-bit OUI with a unique 24-bit value that they generate to create a unique 48-bit address for each Ethernet interface they build. The latter 24-bit value is normally issued sequentally.

Organizationally unique identifier (OUI)/‘company_id’

An OUI/‘company_id’ is a 24 bit globally unique assigned number referenced by various standards. OUI is used in the family of the IEEEE802 LAN standards, e.g., Ethernet, Token Ring, etc.

Standards involved with OUI

The OUI defined in the IEEE 802 standard can be used to generate 48-bit universal LAN MAC addresses to identify LAN and MAN stations uniquely, and protocol identifiers to identify public and private protocols. These are used in local and metropolitan area network applications.

The relevant standards include CSMA/CD (IEEE 802.3), Token Bus (IEEE 802.4), Token Ring (IEEE 802.5), IEEE 802.6, and FDDI (ISO 9314-2).

Structure of the MAC addresses

A MAC address is a sequence of six octets. The first three take the values of the three octets of the OUI in order. The last three octets are administered by the vendor. For

example, the OUI AC - DE - 48 could be used to generate the address AC-DE-48-00-00-80



Address administration

An OUI assignment allows the assignee to generate approximately 16 million addresses, by varying the last three octets. The IEEE normally does not assign another OUI to the assignee until the latter has consumed more than 90% of this block of potential addresses. It is incumbent upon the assignee to ensure that large portions of the address block are not left unused in manufacturing facilities.

9.3.4 Destination address

The destination address field of 48 bits follows the preamble. Each Ethernet interface has a unique 48-bit MAC address. This is a physical or hardware address of the interface that corresponds to the address in the destination address field. The field may contain a multicast address or a standard broadcast address.

Each Ethernet interface on the network reads each frame at least up to the end of the destination address field. If the address in the field does not match its own address, then the frame is not read further and is discarded by the interface.

A destination address of all ‘1’s (FF-FF-FF-FF-FF-FF) means that it is a broadcast and that the frame is to be read by all interfaces.

Destination address in DIX standard

The first bit of the address is used to distinguish unicast from multicast/broadcast addresses. If the first bit in the field is zero, then the address is taken to be a physical (unicast) address. If first bit is one, then the address is taken to be multicast address, meaning that the frame is being sent to several (but not all) interfaces.

Destination address in IEEE 802.3

Here, apart from the first bit being significant as in the DIX standard, the second bit is also significant. If the first bit is zero and the second bit is set to zero as well, then the address in the field is a globally administered physical address (assigned by manufacturer of the interface). If first bit is zero but second bit is set to one, then the address is locally administered (by the systems designer/administrator). The latter option is very rarely used.

9.3.5 Source address

This is the next field in the frame after destination address field. It contains the physical address of the interface of transmitting station. This field is not interpreted in any way by the Ethernet MAC protocol, but is provided for use of higher protocols.

Source address field in DIX standard

The DIX standard allows for changing the source address, but the physical address is commonly used.

Source address in IEEE 802.3 standard

IEEE 802.3 does not provide specifically for overriding 48-bit physical addresses assigned by manufacturers, but all interfaces allow for overriding if required by the network administrator.

9.3.6 Type/length field

This field refers to the data field of the frame.

Type field in DIX standard

In the DIX standard, this field describes the type of high-level PDU information that is carried in the data field of the Ethernet frame. For example, the value of 0x0800 (0800 Hex) indicates that the frame is used to carry an IP packet.

Length/type field in IEEE 802.3 standard

When the IEEE 802.3 standard was first introduced, this field was called the length field, indicating length (in bytes) of data to follow. Later on, (in 1997) the standard was revised to include either a type specification or a length specification.

The length field indicates how many bytes are present in data field, from a minimum of zero to a maximum of 1500.

The most important reason for having a minimum length frame is to ensure that all transmitting stations can detect collisions (by comparing what they transmit with what they hear on the network) while they are still transmitting. To ensure this all frames must transmit for more than twice the time it takes a frame to reach the other end.

The data field must contain a minimum 46 bytes or a maximum of 1500 bytes of actual data. The network protocol itself is expected to provide at least 46 bytes of data. If data is less than 46 bytes, padding by dummy data is done to bring the field size to 46 bytes. Before the data in the frame is read, the receiving station must know which of the bytes constitute real data and which part is padding. Upon reception of the frame, the length field is used to determine the length of valid data in the data field, and the pad data is discarded.

If the value in the length/type field is numerically less than or equal to 1500, then the field is being used as a length field, in which case the number in this field represents the number of data bytes in the data field.

If the value in the field is numerically equal to or greater than 1536 (0x0600), then the field is being used as type field, in which case the hexadecimal identifier in the field is used to indicate the type of protocol data being carried in the data field.

9.3.7 Use of type field for protocol identification

The type field number, issued by the IEEE Type Field Registrar, provides a context for interpretation of the data field of the frame. Well-known protocols have defined type numbers.

The IEEE 802.3, Length/type field, originally known as EtherType, is a two-octet field, which takes one of two meanings depending on its numeric value. For numeric evaluation, the first octet is the most significant octet of this field.

When the value of this field is greater than or equal to 1536 decimal, (0x0600) the type field indicates the nature of the MAC client protocol (type interpretation). The length and type interpretations of this field are mutually exclusive.

The type field is very small and therefore its assignment is limited. It is incumbent upon the assignee to ensure that requests for type fields be very limited and only on an as needed basis. Requests for multiple type fields by the same applicant are not granted unless the applicant certifies that they are for unrelated purposes. In particular, only one new type field is necessary to limit reception of a new protocol or protocol family to the intended class of devices. New protocols and protocol families should have provision for a sub type field within their specification to handle different aspects of the application (e.g., control vs. data) and future upgrades.

9.3.8 Data field

Data field in the DIX standard

In the DIX standard, the data field must contain a minimum of 46 bytes and a maximum of 1500 bytes of data. The network protocol software provides at least 46 bytes of data if needed.

Data field in the IEEE 802.3 standard

Here the minimum and the maximum lengths are same as for the DIX standard. The LLC protocol as per IEEE 802.2 may occupy some space in the data field for identifying the type of protocol data being carried by the frame if the type/length field is used for length information. The LLC PDU is carried in the first 10 bytes in the data field.

If the number of LLC octets is less than the minimum number required for the data filed, then pad data octets are automatically added. On receipt of the frame, the length of meaningful data is determined using the length field.

9.3.9 Frame check sequence (FCS) field

This last field in a frame, the same in both DIX and IEEE 802.3 standards, is used to check the integrity of bits in various fields, excluding, of course, the preamble and FCS fields. The CRC (cyclic redundancy check) method is used to compute and check this integrity.

9.4 LLC frames and multiplexing

9.4.1 LLC frames

Since some networks use the IEEE 802.2 LLC standard, it will be useful to examine LLC frames.

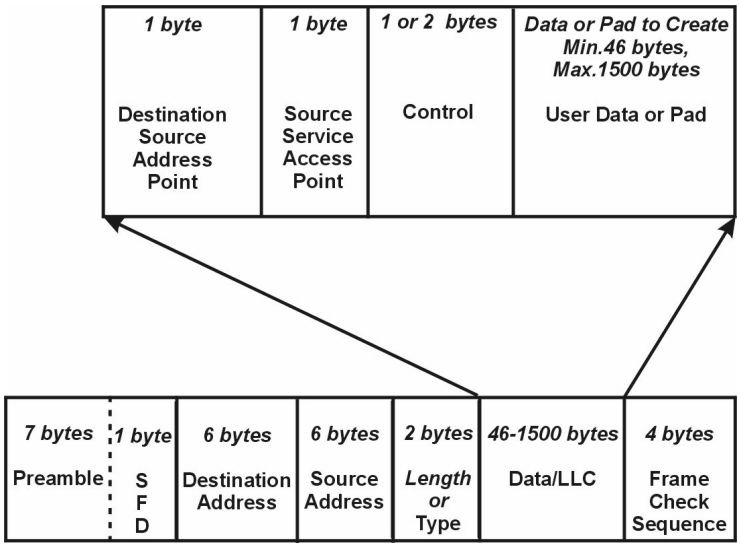


Figure 9.2
LLC protocol data as part of Ethernet frame

9.4.2 LLC and multiplexing

It previous paragraphs it was seen shown that the value of the identifier in the length/type field determines how this field is used. When used as a length field, the IEEE 802.2 LLC identifies the type of high-level protocol being carried in the data field of the frame. The IEEE 802.2 PDU contains a destination service access point or DSAP, (this identifies the high level protocol that the data in the frame is intended for), a source service access point, or SSAP (this identifies the high level protocol from which the data in the frame originated), some control data, and actual user data. Multiplexing and de-multiplexing work in the same way that they do for a frame with a type field. The only difference is that identification of the type of high-level protocol data is shifted to the SSAP, which is located in the LLC PDU. In frames carrying LLC fields, the actual amount of data that can be carried is 3-4 bytes less than in frames that use a type field because of the size of the LLC header.

The reason why the IEEE defined the IEEE 802.2 LLC protocol to provide multiplexing, when the type field does the job equally well, was its objective of standardizing a whole set of LAN technologies and not just IEEE 802.3 Ethernet systems.

802.1p/Q VLAN standards and frames

The IEEE 802.1D standard lays down norms for bridges and switches. The IEEE 802.1p standard, which is a part of 802.1D, provides support for traffic-class expediting and dynamic multicast filtering. It also provides a generic attribute registration protocol (GARP) that can be used by switches and stations to exchange information about various capabilities or attributes that may apply to a switch or port.

The IEEE 802.1Q standard provides a vendor-independent way of implementing VLANs. A VLAN is a group of switch ports that behave as if they are independent switching hub. Provision of VLAN capabilities on a switching hub enables a network manager to allocate a particular set of switch ports to different VLANs.

VLANs can now be based on the content of frames instead of just ports on the hub. There are proprietary frame tagging mechanisms for identifying or tagging frames in terms of which VLAN they belong to. The IEEE 802.1Q provides a vendor independent way of tagging Ethernet frames and thereby implementing VLANs.

Here, 4 bytes of new information containing identification of the protocol and priority information are added after the source address field and before the length/type field. The VLAN tag header is added to the IEEE 802.3 frame as shown below in Figure 9.3:

The tag header consists of two parts. The first part is the tag protocol identifier (TPID), which is a 2-byte field that identifies the frame as a tagged frame. For Ethernet, the value of this field is 0x8100.

The second part is the tag control information (TCI), which is a 2-byte field. The first three bits of this field are used to carry priority information based on the values defined in the IEEE 802.1p standard. The last eleven bits carry VLAN identifier (VID) that uniquely identifies the VLAN to which the frame belongs.

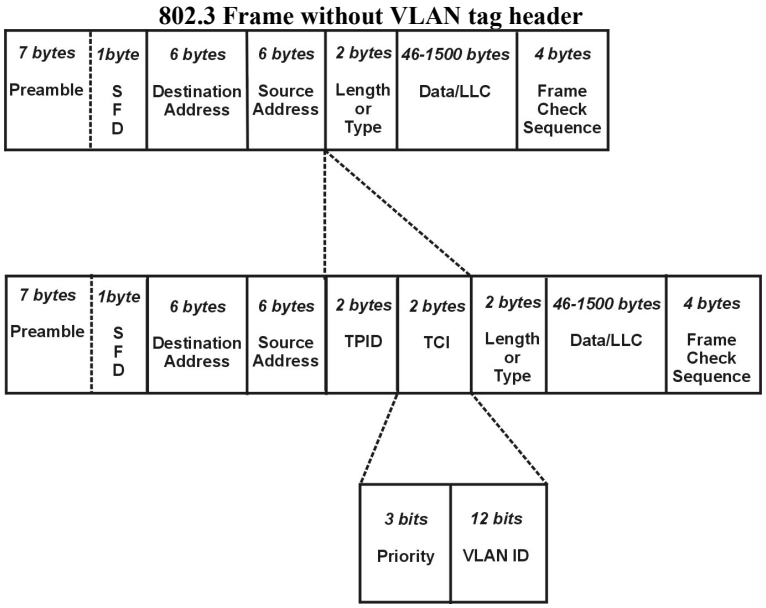


Figure 9.3
IEEE 802.1p/Q VLAN Tag header added to IEEE 802.3 frame

802.1Q thus extends the priority-handling aspects of the IEEE 802.1p standard by providing space in the VLAN tag to indicate traffic priorities.

Addition of the VLAN tag increases the maximum frame size to 1522 bytes.

9.5 Media access control for half-duplex LANs (CSMA/CD)

The MAC protocol for half-duplex Ethernets (DIX as well as IEEE 802.3), decides how to share a single channel for communication between all stations, the transmission being in both directions on the same channel, but not simultaneously.

There is no central controller on the LAN to decide about these matters. Each interface of each station has this protocol and plays by the same MAC rules so that there is a fair sharing of the communication on the channel.

The MAC protocol for half-duplex Ethernet LANs is called CSMA/CD, or carrier sense multiple access/collision detection, after the manner in which the protocol manages the communication traffic.

A detailed look will now be taken at this mechanism.

9.5.1 Terminology of CSMA/CD

Before one gets into a discussion on CSMA/CD, it is necessary to understand the terminology used to describe various features and occurrences of signal transmission in half-duplex Ethernet:

When a message is in the process of being transmitted, the condition is called **carrier**. There is no real carrier signal involved as Ethernet uses a baseband mechanism for transmitting information.

When there is no carrier, the channel is said to be **idle.**

If the channel is not idle, a station wanting to transmit waits for the channel to become idle. This waiting is called **deferring.**

When the channel becomes idle, a station wanting to transmit waits for a predetermined period of time called the **interframe gap.**

When two (or more) signals traveling in opposite directions meet, they **collide** and obstruct each other.

When a transmitting station comes to know of such a collision, the station stops transmission and reschedules it. This is called **collision-detect.**

When collision-detect has taken place, the transmitting station will still transmit 32 bits of data. This data is called a **collision enforcement jam signal** or just **jam signal.** A jam signal is a signal composed of alternating ones and zeroes

After sending a jam signal the transmitting station waits for a random time period before it attempts to transmit again. This waiting is called **back-off.**

The maximum round trip time (RTT) for signal transmission on a LAN (time to go to the farthest station and come back) is called the **slot time.**

9.5.2 CSMA/CD access mechanism

The way the CSMA/CD access mechanism operates is described below:

- A station wishing to transmit listens for the absence of carrier in order to determine if the channel is idle
- If the channel is idle, once the period of inactivity has equaled or exceeded the interframe gap (IFG), the station starts transmitting a frame immediately. If multiple frames are to be transmitted, the station waits for a period of IFG between each successive frame

The IFG is meant to provide recovery time for the interfaces and is equal to the time for the transmission of 96 bits. This is equal to 9.6 microseconds (1 microsecond = 10^{-6} second) for a 10 Mbps network, and 960 nanoseconds and 96 nanoseconds for 100 Mbps and Gigabit networks respectively. 1 nanosecond equals 10^{-9} second.

- If there is a carrier, the station continuously defers till the channel becomes free
- If, after starting transmission, a collision is detected, the station will send another 32 bits of a jam signal. If collision is detected very early or just after the start of transmission, the station will transmit a preamble plus jam signal
- As soon as a collision is detected, two processes are invoked. These are a collision counter and a back-off time calculation algorithm (which is based on random number generation)
- After sending the jam signal, the station stops transmitting and waits for a period equal to the back-off time (which is calculated by the aforementioned

algorithm). On expiry of the back-off time, the station starts transmitting all over again

- If a collision is detected once again, the whole process of backing off and re-transmitting is repeated, but the algorithm, which is given the collision count by the counter, increases the back-off time. This can go on till there are no collisions detected, or upto a maximum of 16 consecutive attempts.
- If the station has managed to send the preamble plus 512 bits, the station has 'acquired' the channel, and will continue to transmit until there is nothing more to transmit. If the network has been designed as per rules, there should be no collisions after acquiring the channel
- The 512-bit slot time mentioned above is for 10 Mbps and 100 Mbps networks. For gigabit networks, this time is increased to 512 bytes (4096 bits).
- On acquiring the channel the collision counter and back-off time calculation algorithm are turned off
- All stations strictly follow the above rules. It is of course assumed that the network is designed as per rules so that all these timing rules provide the intended results

9.5.3 Slot time, minimum frame length, and network diameter

Slot time is based on the maximum round trip signal traveling time of a network. It includes time for a frame to go through all cable segments and devices such as cables, transceivers and repeaters along the longest route.

Slot time has two constituents:

- The time to travel across a maximum sized system from end to end and return
- The maximum time for collision detection, and sending of the jam signal

These two times plus a few bits for safety amount to the slot time of 512-bits for 10 Mbps and 100 Mbps systems. Thus, even when transmitting the smallest legitimate frame, the transmitting station will always get enough time to know about a collision even if collision occurs at the farthest end of the longest route.

The minimum frame size of 512 bits includes 12 bytes for addresses, 2 bytes for length/type field, 46 bytes of data and 4 bytes of FCS. Preamble is not included in this calculation.

The slot time and network size are closely related. This is a trade-off between the smallest frame size and the maximum cable length along the longest route. The signal speed is dependent on the medium – around two-thirds the speed of light on copper, regardless of the bit rate. The transmission speed, for instance 10 Mbps, determines the LENGTH of the frame (in time). Therefore, when going from 10 Mbps to 100 Mbps (factor of 10), the maximum frame size in microseconds will 'shrink' by a factor of 10, and hence the permissible collision domain will reduce from 2500 m. to 250 m. The slot time will reduce from 512 microseconds to 5.12 microseconds. For Gigabit Ethernet the same argument would produce a collision domain of 25 m. and a slot time of 0.512 microseconds, which is ridiculously low. Therefore, the minimum frame size for the latter has been increased to 512 bytes (4096 bits), which gives a physical collision domain of around 200 m. The slot time is used as a basic unit of time for the back-off time calculation algorithm.

In practice many fast Ethernet systems and most Gigabit Ethernet systems operate in full-duplex mode, i.e. with the collision detection circuitry disabled. However, all these

systems have to conform to the collision detection requirements for backwards compatibility with the original IEEE 802.3 standard.

Since a valid collision can occur only within the 512-bit slot time, the length of a frame destroyed in a collision, a 'fragment', will always be smaller than 512 bits. This helps interfaces in detecting fragments and discarding them.

9.5.4 Collisions

Collisions are a normal occurrence in CSMA/CD systems. Collisions are not errors, and they are managed efficiently by the protocol. In a properly designed network, collisions, if they occur, will happen in the first 512 bits of transmission. Any frame encountering a collision is destroyed, its fragments are not bigger than 512 bits. Such a frame is automatically retransmitted without fuss. The number of collisions will increase with transmission traffic.

Collision detection mechanisms

The mechanism for detection of collision depends on the medium of transmission.

On a coaxial cable the transceiver detects collisions by monitoring the average DC signal voltage level which reaches a particular level, triggering the collision detect mechanism circuit.

On link segments, such as twisted-pair or fiber optic media, (which have independent transmit and receive data paths) collisions are detected in a link segment receiver by simultaneous occurrence of activity on both transmit and receive data paths.

Late collisions

A collision in a smoothly functioning network has to occur within the first 512-bit time. If a late collision occurs it signifies a serious error. There is no automatic retransmission of a frame in case of a late collision occurring, and the fault has to be detected by higher-level protocols. The sending interface must wait for acknowledge timers to time-out before resending the frame. This slows down the network.

Late collisions are caused by network segments that exceed the stipulated maximum sizes, or by a mismatch between duplex configurations at each end of a link segment. One end of a link segment may be configured for half-duplex transmission while the other end maybe configured for full-duplex transmission (full-duplex does not use CSMA/CD for obvious reasons).

Excessive signal crosstalk on a twisted pair segment can also result in late collisions.

Collision domains

A collision domain is a single half-duplex Ethernet system of cables; repeaters, station interfaces, and other network hardware where all are a part of the same signal-timing and slot-timing domain. A single collision domain may encompass several segments as long as they are linked together by repeaters. (A repeater enforces any collisions on any segment attached to it, for example, a collision on segment x is enforced by the repeater onto segment y by sending a jam signal.) A repeater makes multiple network segments function like a single cable.

9.6 MAC (CSMA-CD) for gigabit half-duplex networks

Most Gigabit Ethernet systems use the full-duplex method of transmission, so the question of collisions does not arise. However, the IEEE has specified a half-duplex CSMA/CD mode for gigabit networks to insure that gigabit networks get included in the IEEE 802.3 standard. This matter is narrated in brief below for the sake of completeness.

If the same norms (same minimum frame length and slot times) are applied to gigabit networks then the effective network diameter becomes very small at about 20 meters. This is too small a value to be of any practical use. A network diameter of say 200 meters would be more useful.

To solve this problem the slot time is increased while keeping the same minimum frame length. The only way this can be done is by increasing the 'minimum frame length'. Just specifying a longer length would make the system incompatible with systems using 512 bits as the minimum frame length. Appending or suffixing non-data signals, called extension bits, at the end of an FCS field, has overcome this problem. This is called 'carrier extension'.

With carrier extension, the minimum frame size is increased from 512 bits (64 bytes) to 4096 bits (512 bytes). This now increases the slot time proportionately. All this, of course, increases overhead i.e. it decreases the proportion of actually useful data (original data) to total traffic, thereby decreasing efficiency of the network.

All this is somewhat academic since most gigabit Ethernets use full-duplex methods so that CSMA/CD is not at all needed.

9.7 Multiplexing and higher level protocols

Several computers using different high-level protocols can use the same Ethernet network. Identifying which protocol is being used and carried in each data frame is called multiplexing, which allows placing of multiple sources of information on a single system.

The type field was originally used for multiplexing, For example, a higher-level protocol creates a packet of data, and software inserts an appropriate hexadecimal value in the type field of the Ethernet frame. The receiving station uses this value in the type field to de-multiplex the received frame.

The most widely used high-level protocol today is TCP/IP, which can use both type and length fields in the Ethernet frame. Newer high-level protocols developed after the creation of the IEEE 802.2 LLC use the length field and LLC mechanism for multiplexing and de-multiplexing of frames.

9.8 Full-duplex transmissions

The full-duplex mode of transmission allows simultaneous two-way communication between two stations, which must use point-to-point links with media such as twisted-pair or fiber optic cables to provide independent transmit and receive data paths. Because of the absence of CSMA/CD there can only be two nodes (for example an NIC and switch port) in a collision domain.

Full-duplex mode doubles the bandwidth of media as compared with that of half-duplex mode. The maximum segment length limitation imposed by timing requirements of half-duplex mode does not apply to full-duplex mode.

The IEEE specifies full-duplex mode in its 802.3x supplement (100 Mbps Ethernet) along with optional mechanisms for flow control, namely MAC Control and PAUSE.

9.8.1 Features of full-duplex operation

For full-duplex operation, certain requirements must be fulfilled. These include independent data paths for transmit and receive mode in cabling media, a point-to-point link between stations, and the capability and configuration of interfaces of both stations for simultaneous transmission and receipt of data frames.

In full-duplex mode a station wishing to transmit ignores carrier sense and does not have to defer to traffic. However, the station still waits for an interframe gap period between its own frame transmissions, as in half-duplex mode, so that interfaces at each end of the link can keep up with the full frame rate.

Since there are no collisions, the CSMA/CD mechanism is deactivated at both ends.

Although full-duplex mode doubles bandwidth, this usually does not result in a doubling of performance because most network protocols are designed to send data and then wait for an acknowledgment. This could lead to heavy traffic in one direction and negligible traffic in the return direction. However, a network backbone system using full-duplex links between switching hubs will typically carry multiple conversations between many computers and the aggregate traffic on a backbone system will therefore tend to be more symmetrical.

It is essential to configure both ends of a communication link for full-duplex operation, or else serious data errors may result. Auto-negotiation for automatic configuration is recommended wherever possible. Since support for auto-negotiation is optional for most media systems, a vendor may not have provided for it. In such case careful manual configuration of BOTH ends of the links is necessary. One end obeying full duplex while the other is still on half-duplex will definitely result in loss of frames.

9.8.2 Ethernet flow control

Network backbone switches connected by full-duplex links can be subject to heavy traffic, sometimes overloading internal switching bandwidth and packet buffers, which are apportioned to switching ports. To prevent overloading of these limited resources, a variety of flow control mechanisms (for example use of a short burst of carrier signal sent by a switching hub to cause stations to stop sending data if buffers are full) are offered by hub vendors for use on half-duplex segments. These are not, however, useful on full-duplex segments.

The IEEE has provided for optional MAC Control and PAUSE specifications in its 802.3x Full-duplex supplement.

9.8.3 MAC control protocol

The MAC control system provides a way for the station to receive a MAC control frame and act upon it. MAC control frames are identified with a type value of 0x8808. A station equipped with optional MAC control receives all frames using normal Ethernet MAC functions, and then passes the frames to the MAC control software for interpretation. If the frame contains hex value 0x8808 in the type field, then the software reads the frame, reads MAC control operation codes carried in the data field and takes action accordingly.

MAC control frames contain operational codes (opcodes) in the first two bytes of the data field.

9.8.4 PAUSE operation

The PAUSE system of flow control uses MAC control frames to carry PAUSE commands. The opcode for the PAUSE command is 0x0001.

When a station issues a PAUSE command, it sends a PAUSE frame to 48-bit MAC address of 01-80-C2-00-00-01. This special multicast address is reserved for use in PAUSE frames, simplifying the flow control process, because a frame with this address will not be forwarded to any other port of the hub, but will be interpreted and acted upon.

The particular multicast address used is selected from a range of addresses that have been reserved by the IEEE 802.1D standard.

A PAUSE frame includes the PAUSE opcode as well as the period of pause time (in the form of a two byte integer) being requested by the sending station. This time is the length of time for which the receiving station will stop transmitting data. Pause time is measured in units of 'quanta' where one quanta is equal to 512 bit times.

Figure 9.4 shows a PAUSE frame wherein the pause time requested is 3 quantas or 1536 bit times.

A PAUSE request provides real-time flow control between switching hubs, or even between a switching hub and a server, provided of course they are equipped with optional MAC control software and are connected by a full-duplex link.

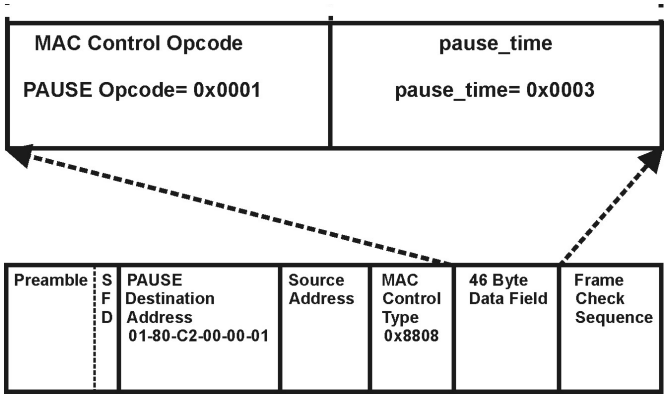


Figure 9.4
PAUSE frame with MAC control opcode and pause time

9.9 Auto-negotiation

9.9.1 Introduction to auto-negotiation

Auto-negotiation involves the exchange of information between stations about their capabilities over a link segment and performing of automatic configuration to achieve the best possible mode of operation over a link. Auto-negotiation enables an Ethernet equipped computer to communicate at the highest speed offered by a multi-speed switching hub port.

A switching hub capable of supporting full-duplex operation on some or all of its ports can announce the fact using auto-negotiation. Stations supporting full-duplex operation and connected to the hub can then automatically configure themselves to use the full-duplex mode when interacting with the hub.

Automatic configuration using auto-negotiation makes it possible to have twisted-pair Ethernet interfaces that can support several speeds. Twisted-pair ports and interfaces can configure themselves to operate at 10 Mbps, 100 Mbps, or 1000 Mbps.

The auto-negotiation system has the following features:

- It is designed to work over link segments only. A link segment can have only two devices connected to it, one at each end
- Auto-negotiation takes place during link initialization. When a device is turned on, or an Ethernet cable is connected, the link is initialized by the devices at each end of the link. This initialization and auto-negotiation happens only once, before transmission of any data over the link

- Auto-negotiation uses its own signaling system. This signaling system is designed for twisted-pair cabling

9.9.2 Signaling in auto-negotiation

Auto-negotiation uses fast link pulse (FLP) signals to carry information. These signals are a modified version of normal link pulse (NLP) signals used for verifying link integrity on 10BaseT system.

FLPs are specified for following twisted-pair media systems:

- 10BaseT
- 100BaseTX (using unshielded twisted-pair)
- 100BaseT4
- 100BaseT2
- 1000BaseT

100BaseTX with shielded twisted-pair cable and 9-pin connectors will not support auto-negotiation. There is also no IEEE auto-negotiation standard for fiber optic Ethernet systems except for fiber optic gigabit Ethernet systems.

9.9.3 FLP details

Fast link pulses are sent in bursts of pulses in 33 pulse positions where each pulse position may contain a pulse. Each pulse is 100 nanoseconds long and the time between each successive burst the same as that between NLPs. This fact deludes a 10BaseT device that is receiving an NLP; thus providing backward compatibility with older 10BaseT equipment that does not support auto-negotiation.

Of the 33 pulse positions, the 17 odd-numbered positions each holds a link pulse that represents clock information. The 16 even-numbered pulse positions carry data. The presence or absence of a pulse in an even numbered position represents logic 1 and logic 0 respectively. This coding in even numbered positions is for the transmission of 16-bit link code words that contain auto-negotiation information.

Each burst of 33 pulse positions contains a 16-bit message, and a device can send as many bursts as are needed. Sometimes the negotiation task may get completed in the first message in the first burst itself. The first message is called the base page. Mapping of the base page message is shown in Figure 9.5.

The 16 bits are labeled D0 through D15. D0 to D4 are used as a selector field for identifying the type of LAN technology in use, allowing the protocol to be extended to other LANs in future. For Ethernet, the S0 bit is set to 1 and S2 to S4 are all set to zero.

The 8-bit field from D5 to D12 is called the technology ability field. Positions A0 to A7 are allotted to the presence or absence of support for various technologies as shown in Figure 9.5. If a device supports one or more technologies, the corresponding bits are set to 1; else they are set to zero as the case may be. Two reserved bit positions, A6 and A7, are for future use.

Bit D13 is called the remote fault indicator. This bit can be set to 1 if a fault has been detected at a remote end.

Bit D14 is set to 1 to acknowledge receipt of the 16-bit message. Negotiation messages are sent repeatedly until the link partner acknowledges them, thus completing the auto-negotiation process.

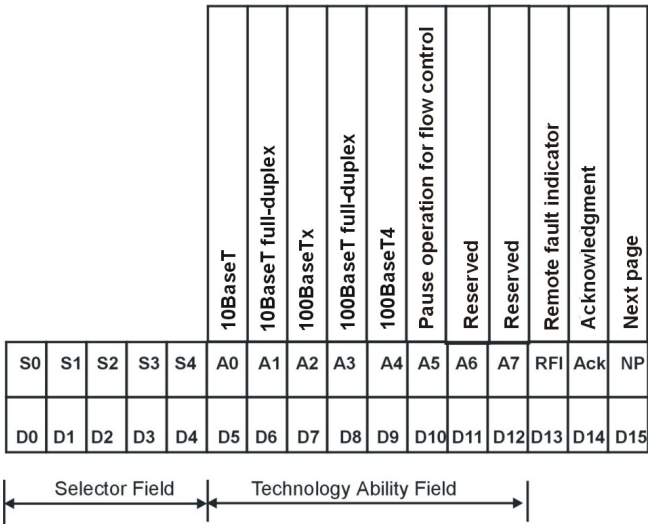


Figure 9.5
Auto-negotiation base page mapping

The link partner sends an acknowledgement only after three consecutive identical messages are received.

Bit D15 is used to indicate if there is a next page, that is, if more information on capabilities is to follow. Next page capability has been provided for sending vendor-specific information or any new configuration commands that may be required in future developments. 1000BaseT gigabit systems use this method for their configuration.

Once auto-negotiation is complete, further bursts of pulses will not be sent unless a link has been down due to any reason and reconnection takes place.

9.9.4 Matching of best capabilities

Once devices connected to two ends of a link have informed each other of their capabilities, the auto-negotiation protocol finds the best match for transmission or the highest common denominator between the capabilities of the two devices. This is based on priorities specified by the standard. Priority is decided by type of technology and not by the order of bits in the technology ability field of base page.

Priorities, ranking from highest to lowest, are as follows:

1	1000BaseT	full duplex
2	1000BaseT	
3	100BaseT2	full duplex
4	100BaseTX	full duplex
5	100BaseT2	
6	100BaseT4	
7	100BaseTX	
8	10BaseT	full duplex
9	10BaseT	

Thus if both devices support say, 100BaseTX as well as 10BaseT, then 100BaseTX will be selected for transmission. Note that other things being the same, full-duplex has higher priority than half-duplex.

If both devices support the PAUSE protocol and the link is configured for full-duplex, then PAUSE control will be selected and used. The priority list above is based on data

rates while PAUSE control has nothing to do with data rates. Therefore, PAUSE is not part of the priority list.

If auto-negotiation does not find any common support on devices at both ends, then a connection will not be made at all and the port will be left in the 'off' position.

9.9.5 Parallel detection

Auto-negotiation is optional for most media systems (except for 1000BaseT systems) because many of the media systems were developed before auto-negotiation was developed. Therefore, the auto-negotiation system has been made compatible with those devices that do not have it. If Auto-negotiation exists only at one end of a link, then the protocol is designed to detect this condition and respond by using 'parallel detection'.

Parallel detection can detect the media system supported at the other end and can hence set the port for that media system. It, however, cannot detect whether the other end supports full-duplex or not. Even if the other end supports full-duplex, parallel detection will set the port for half-duplex mode. Parallel detection is not without problems, and it is preferable for network managers to go for auto-negotiation on all their devices.

9.10 Deterministic Ethernet

Deployment of Ethernet in the industrial environment is increasing but a debate goes on as to whether Ethernet can deliver where mission critical applications are concerned. The issue here is the requirement of a deterministic system when industrial process controls are the application environment

Many people look down upon Ethernet because the CSMA/CD systems are not deterministic. A deterministic system in this context means a system that will deliver the transmission in very short time – as specified and required by the process control parameter. The CSMA/CD style of operation of Ethernet is definitely not deterministic, but probabilistic. However, this lack of determinism is being steadily chiseled away.

The features that are making Ethernet deterministic are as follows:

- Full-duplex operation is deterministic, because CSMA/CD is irrelevant here
- VLANs and prioritized frames (IEEE 802.1p/Q) have also moved Ethernet towards full determinism

More on this subject and on industrial Ethernet will follow later in this manual.